SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE CODE : CSA40 Name: Y.Tharun

COURSE NAME : MANAGEMENT INFORMATION SYSTEM Reg N0: 192110204

2) Draw a coffe coffe day ordering system. A coffe coffee day shop vending machine dispenses coffee to customers. Customers order coffee by selecting a recipe from a set of recipes. Customers pay for the coffee using coins. Change is given back, if any, to the customers. The 'service assistant' loads ingredients (coffee powder, milk, sugar, water, chocolate) into the coffee machine. The 'service assistant' adds recipe by indicating the name of the coffee, the units of coffee powder, milk, sugar, water, chocolate to be added as well as the cost of the coffee. The service assistant can also edit and delete a recipe. Develop the use case diagram for the specification above**.**

AIM: To design a Use Case Diagram for the Coffee Vending Machine Ordering System that illustrates interactions between customers, service assistants, and the coffee machine for ordering and managing coffee recipes.

Procedure:

1. Customer Selects Coffee
    - The customer chooses a coffee type from the available recipes (e.g., Espresso, Cappuccino, Latte).
2. Customer Makes Payment
    - The customer inserts coins into the vending machine to pay for the selected coffee.
3. Machine Verifies Payment
    - The vending machine checks if the inserted amount is sufficient for the selected coffee.
    - If the amount is less, the machine prompts the customer to add more.
    - If the amount is more, the machine calculates the change to be returned.
4. Machine Prepares Coffee
    - The machine dispenses the required ingredients (coffee powder, milk, sugar, water, chocolate) as per the recipe.
5. Coffee is Dispensed
    - The prepared coffee is poured into a cup and made available for the customer.
6. Change is Returned (if applicable)

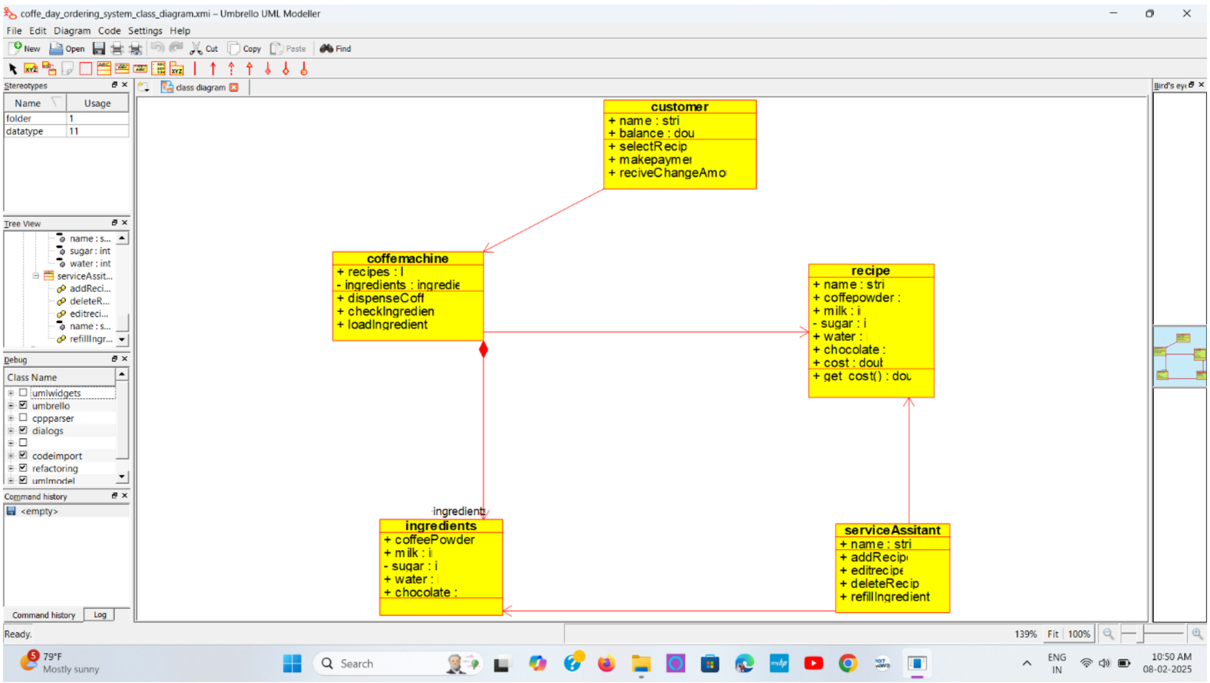
- If extra money was inserted, the machine returns the remaining balance to the customer.

7. Service Assistant Loads Ingredients
   - The assistant refills ingredients (coffee powder, milk, sugar, water, chocolate) to keep the machine operational.
8. Service Assistant Adds or Modifies Recipes
   - The assistant can add new coffee recipes, modify existing ones, or delete old recipes as needed.
9. Machine Displays Status
   - The machine shows the availability of ingredients, coffee recipes, and any maintenance messages.
10. System Maintenance (if needed)

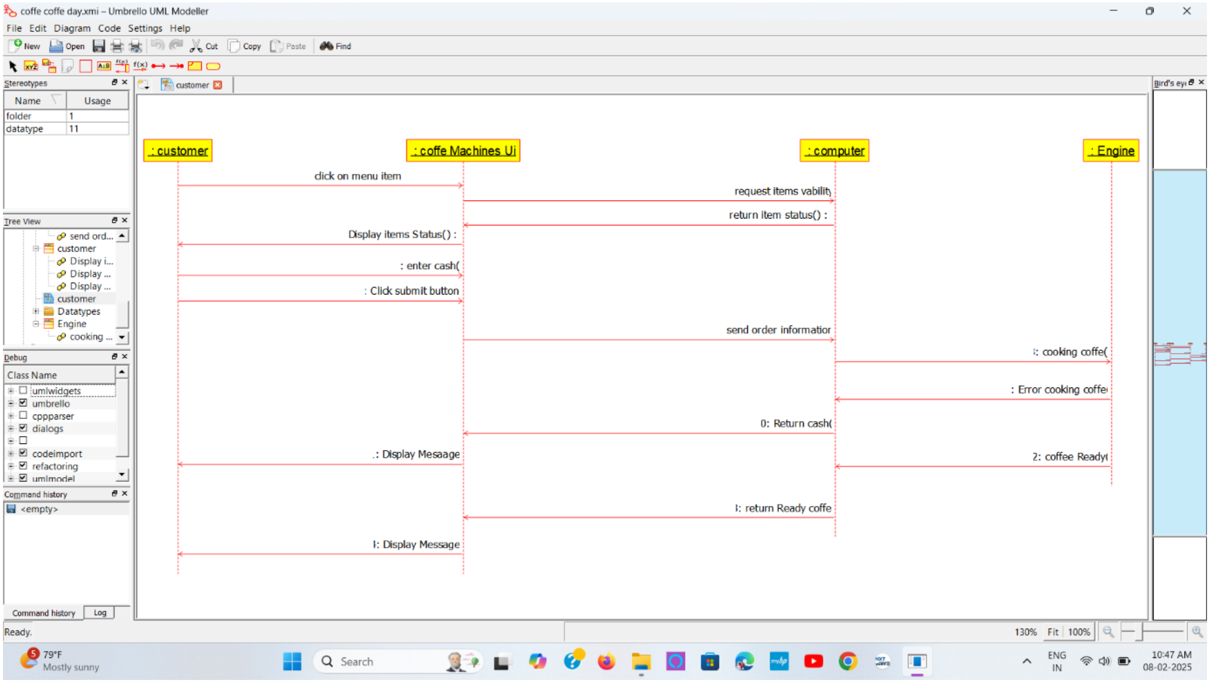
- If the machine encounters any issues (e.g., ingredient shortage, technical error), the service assistant performs maintenance or troubleshooting.

OBSERAVATION

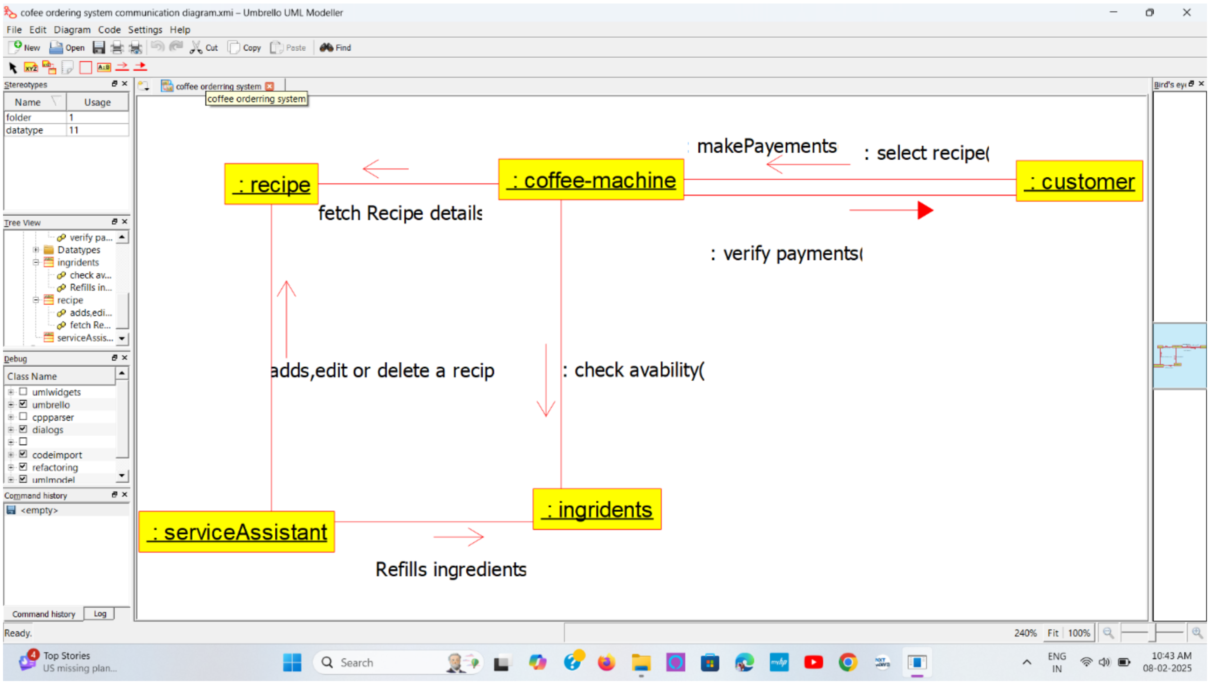
Class Diagram...



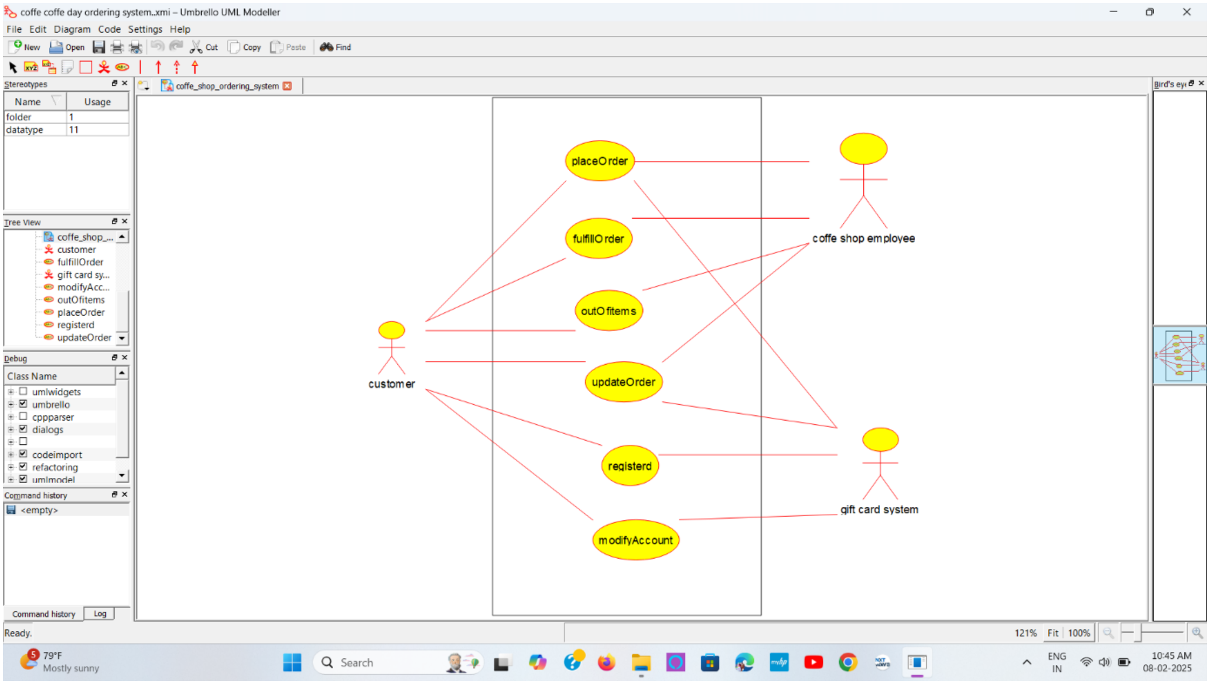
Sequence Diagram...



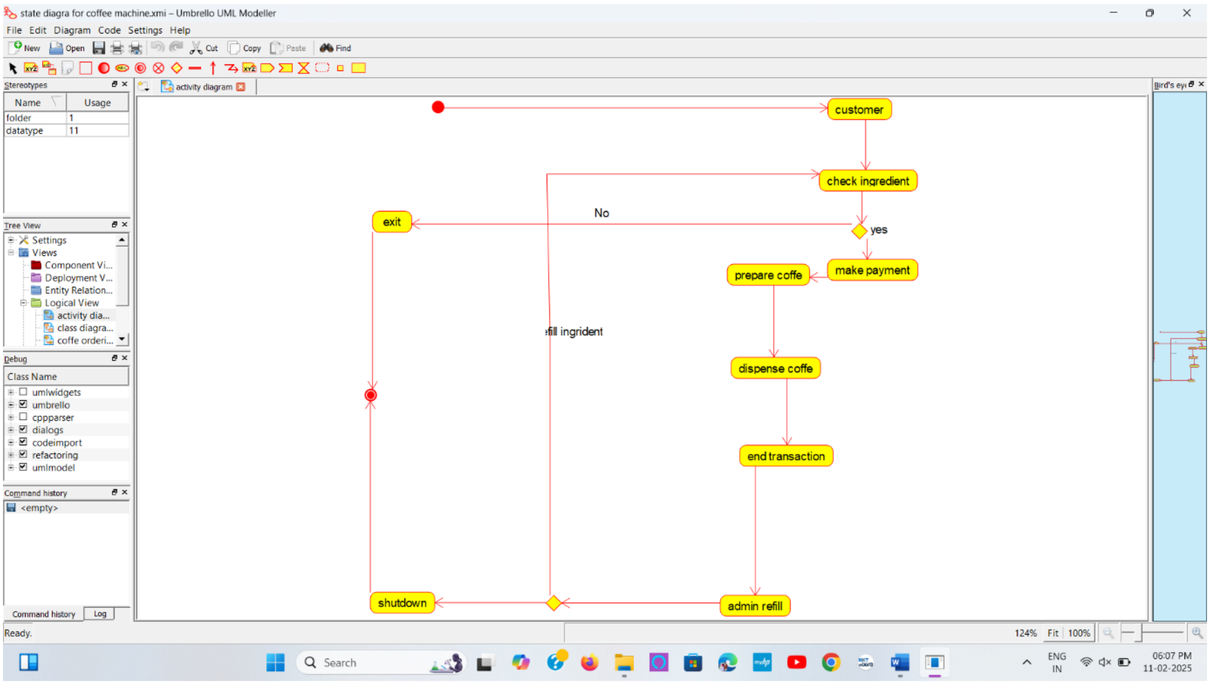
Communication Diagram...



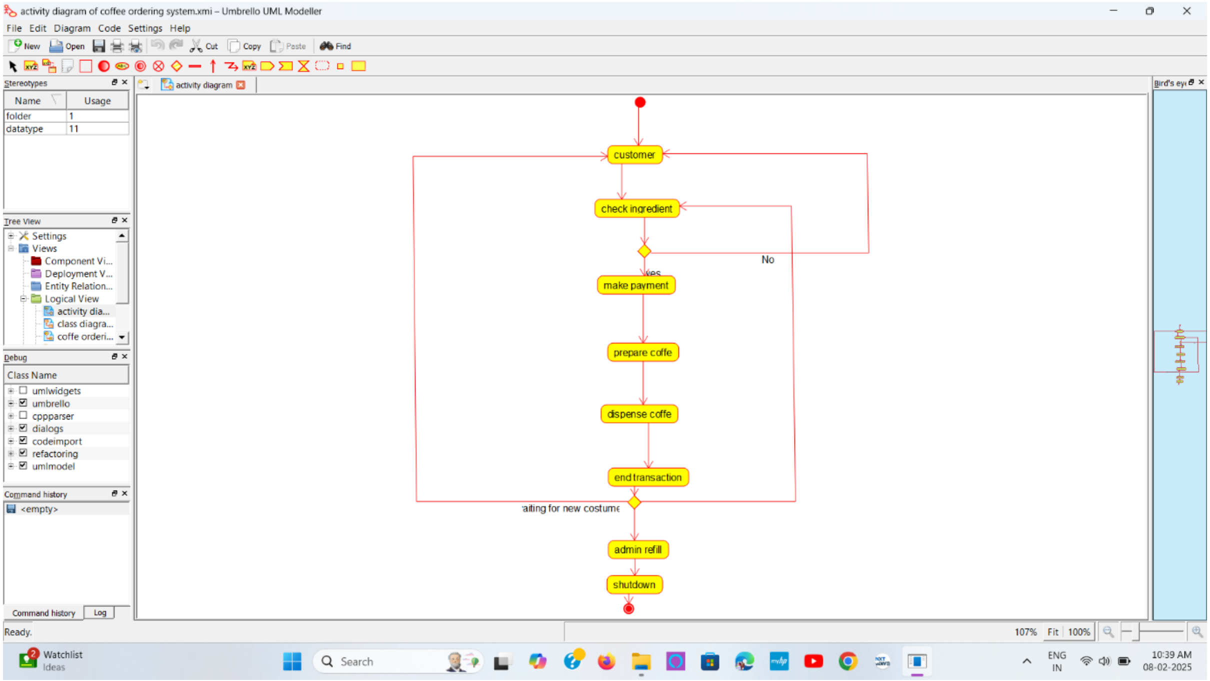
Use Case Diagram...



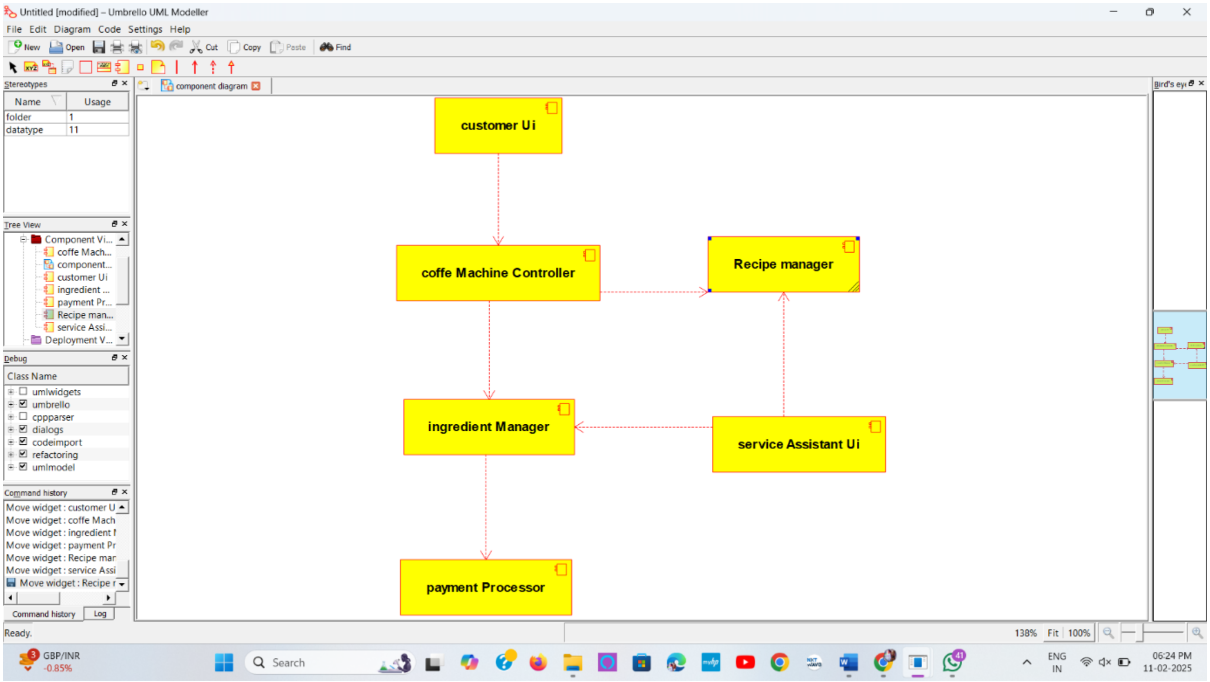
State Diagram...



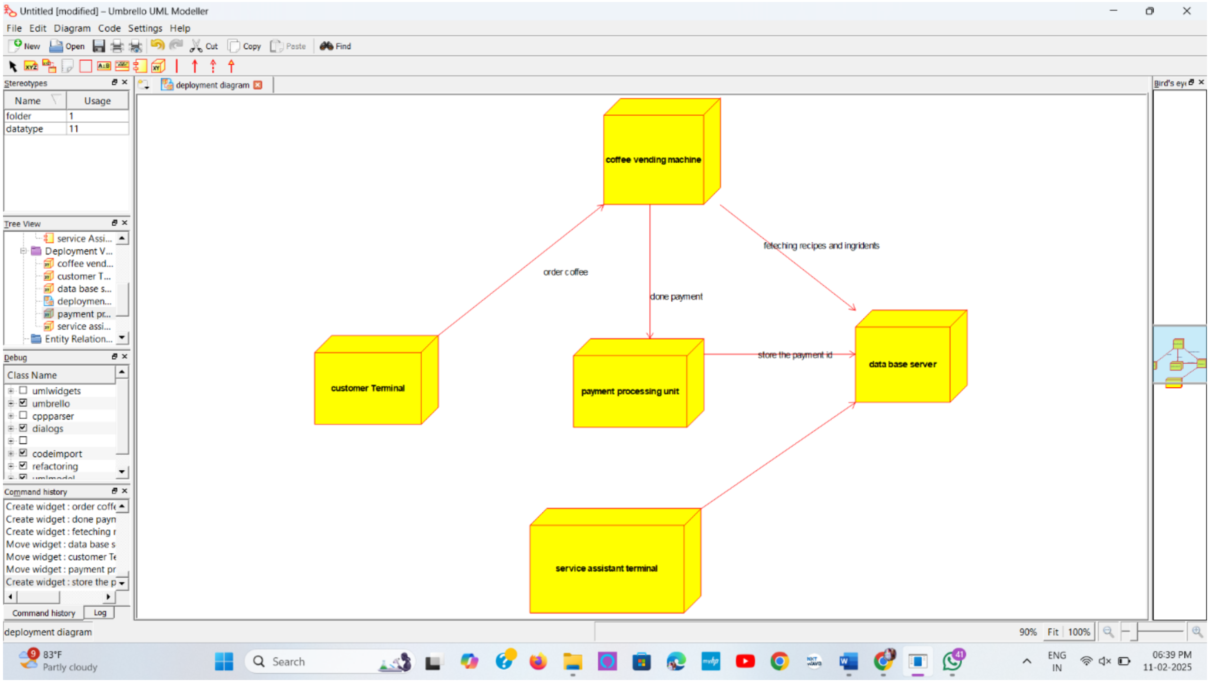
Activity Diagram...



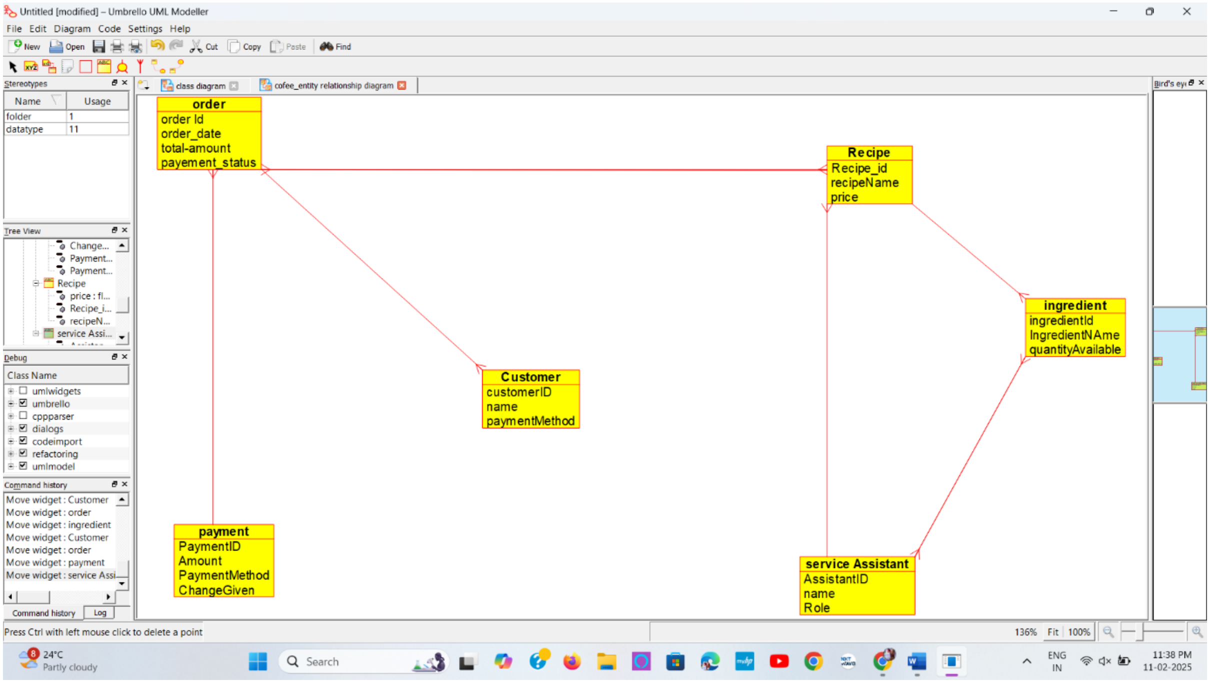
Component Diagram...



Deployment Diagram...



Entity Relationship Diagram...



Result:

The**UML Diagram** for the Coffee Day Ordering System successfully represents the relationships between key entities such as **Customer, Order, Recipe, Ingredient, Service Assistant, and Payment**. This model ensures efficient management of orders, ingredients, and transactions, improving the automation of the coffee vending process.